

Cross Attention and Intra-Layer Attention in Heterogeneous Graph Neural Networks for Drug–Target Interaction Prediction

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Abstract—In recent years, computational prediction of drug–target interactions (DTIs) has become essential for drug discovery and repositioning. However, traditional experimental approaches for DTI identification are time-consuming and costly. To address this, many machine learning-based methods have been developed, yet most existing models neglect important information interaction between drugs and targets in drug–target pairs (DTPs) during drug–target interaction. In this study, we propose a novel cross-attention and intra-layer attention mechanism within a heterogeneous graph neural network (CAIHGNN) for DTI prediction. The cross-attention mechanism allows for dynamic learning of feature correlations between drugs and targets, while the intra-layer attention captures both explicit and implicit interactions within DTPs. Additionally, we introduce a drug–target pair correlation graph to exploit high-order interactions between DTPs. Extensive experiments on two biological heterogeneous datasets demonstrate the superior performance of our proposed method in accurately predicting DTIs. Furthermore, the model exhibits robust generalization in case study, showing promise for real-world drug discovery applications.

Index Terms—Cross attention, intra-layer attention, meta-path.

I. INTRODUCTION

PREDICTION of drug–target interactions (DTIs) plays a crucial role in drug discovery (i.e., identifying new drug

candidates) and drug repositioning (i.e., discovering new indications for existing drugs). Traditional experimental methods for identifying DTIs are time-consuming, expensive, and often impractical for large-scale screening. As a result, machine learning-based approaches have gained significant attention in both academia and industry, offering more cost-effective and scalable solutions. These methods rely heavily on various types of data, including chemical and genomic features of drugs and targets. However, despite their success, many of these methods are primarily focused on using chemical and genomic data, often overlooking other valuable sources of information such as pharmacological and phenotypic data, including disease profiles and side effects. As a result, there is a need for more comprehensive approaches that can integrate these diverse data sources.

To address this, several methods have been proposed for DTI prediction, ranging from traditional similarity-based approaches to more advanced machine learning and deep learning models. These approaches, however, have different strengths and limitations in handling diverse data types and extracting meaningful patterns from them.

Similarity-based methods like DTi2Vec [1] leverage the similarities between drugs and targets by mapping them into low-dimensional vector spaces using algorithms such as Node2Vec, capturing the structural similarities between them. While effective in certain scenarios, these methods often fail to generalize to novel drug–target interactions, especially when data is sparse or incomplete. [2], [3]

Machine learning approaches often rely on chemical and genomic data to build predictive models. Nagamine and Sakakibara [4] proposed a method that combines chemical structure and protein sequence information to predict DTIs. While these models are valuable, they are heavily dependent on feature engineering and may not fully capture the complexities of drug–target relationships.

Deep learning methods [5], [6], [7], such as DeepDTA [8] and DeepConv-DTI [9], utilize convolutional and recurrent neural networks to automatically extract molecular features from raw data. These models have shown impressive performance, but they often require large, high-quality datasets to train effectively, and can be difficult to interpret due to their black-box nature.

However, predicting DTIs goes beyond simple molecular feature extraction, as the relationships between drugs, targets, diseases, and side effects can be much more complex. For instance, a drug’s interaction with a target can lead to off-target

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effects, which might cause unintended therapeutic effects or side effects. These off-target effects, such as minoxidil being repurposed for hair loss or sildenafil (Viagra) for erectile dysfunction, reflect the broader, more complex biological context in which drugs operate [10]. The ability to capture such heterogeneous relationships—not just between drugs and targets, but also including diseases and side effects—could significantly improve DTI prediction. The inherent complexity of these relationships suggests that heterogeneous graph-based methods, which can model the multiple types of interactions between drugs, targets, and other entities (e.g., diseases, side effects), could be a more comprehensive approach. Such models leverage the power of graph representations to integrate diverse data types, enabling the capture of richer, more nuanced patterns that go beyond traditional molecular or genomic features. Heterogeneous graph-based methods have shown promise in capturing complex DTI relationships, and recent advancements have leveraged the superiority of graph neural networks (GNNs) in graph analysis [11], [12]. Many methods based on GCN have been proposed for DTI prediction [13], [14], which model drug-target interactions as a graph and learn node representations to predict relationships. Recent advancements include H2GnnDTI [15], which employs hierarchical heterogeneous graph neural networks to integrate multi-level drug and protein structures, iNGNN-DTI [16], which leverages pre-trained molecule models and nested GNNs to capture substructure interactions, and GeNNius [17], which achieves ultrafast DTI inference through efficient graph neural networks. Additionally, MLGANN [18] uses multi-layer graph attention networks to integrate multi-source data, while SHGNN-DTI [19] models signed heterogeneous graphs to capture both positive and negative DTI effects. Despite their success, these methods face several limitations: most of them learn representations for drugs and targets separately and rely on simplistic aggregation (e.g., summation or averaging), overlooking the nuanced interactions between drugs and targets. Moreover, they often focus on explicit relationships between drug-target pairs, neglecting implicit relationships such as indirect associations through diseases or side effects [20], [21], [22].

To address these challenges, we propose CAIHGNN, a novel GNN-based framework that leverages attention mechanisms to enhance DTI prediction. CAIHGNN integrates a cross-attention mechanism to dynamically model the interactions between drug and target features, overcoming the simplistic aggregation issue. Additionally, it employs an intra-layer attention mechanism to capture implicit relationships between drug-target pairs, followed by a Graph Convolutional Network (GCN) for classification. The technical details of these mechanisms are provided in Section III. The main contributions of this paper are as follows:

- An innovative bidirectional cross attention learning approach is proposed. This bidirectional cross-attention learning mechanism can capture the interaction features between drugs and targets, thereby enhancing the performance of DTI prediction.
- A novel GCN combined with an intra-layer attention network is introduced, which learns both implicit and

explicit information between drug-target pairs, capturing more accurate system-level information to enhance the performance of DTI prediction.

- We conducted extensive experiments to validate the accuracy and effectiveness of the proposed model. Specifically, we performed attention analysis and ablation studies on the Cross Attention and Intra-layer Attention modules to assess their individual contributions and demonstrate the rationale behind their design.

II. PRELIMINARY

This section presents formal definitions of the terms used in this paper.

Definition 1 (Heterogeneous graph): A heterogeneous graph is defined as a directed graph $G = (V, E, A, R)$, where V , E , A , R represent node set, edge set, node type set and edge type set, respectively, and $|A| + |R| > 2$.

Definition 2 (Metapath): A metapath Φ is defined as a path in the form of $\Phi = A_1 \xrightarrow{R_1} A_2 \xrightarrow{R_2} \dots \xrightarrow{R_l} A_{l+1}$ (abbreviated as $A_1 A_2 \dots A_{l+1}$), which describes a composite relation $R = R_1 \circ R_2 \circ \dots \circ R_l$ between node types A_l and A_{l+1} , where $\circ$ denotes the composition operator on relations.

Different metapaths imply different semantics. Particularly, DID indicates a disease can be treated by two drugs, while DTD suggests that two drugs share the same target.

Definition 3 (Metapath instance): Given a metapath Φ , there exist multiple metapath instances ϕ for each node following the schema of Φ .

Definition 4 (Metapath-based neighbors): Given a metapath Φ , the metapath-based neighbors N_v^Φ of node v is defined as the set of nodes connecting to v via metapath instances of Φ . A neighbor node connected by two different metapath instances is regarded as two different nodes in N_v^Φ . Also, N_v^Φ consists of v itself.

Definition 5 (Metagraph): Given a metapath Φ in a heterogeneous graph G , the metagraph G_v^Φ of node v is a subgraph of G , and it is built by all the metapath-based neighbors $u \in N_v^\Phi$, where the intermediate nodes along metapath instances are converted to edge information.

III. METHOD

In this section, we present the details of CAIHGNN. It includes four key building blocks: the drug/target representation learning module, the DTP fusion module, DTP representation learning module and the drug-target interaction prediction module. The overall framework of CAIHGNN is illustrated in Fig. 1.

A. Metapath Construction

These heterogeneous biological data were collected from various public database resources [23], [24], [25], [26], [27]. Based on this data, we constructed a heterogeneous biological graph that includes nodes including drugs, targets, diseases, and side effects. We employ a dual-channel approach to learn the representations of drugs and targets. Specifically, we construct

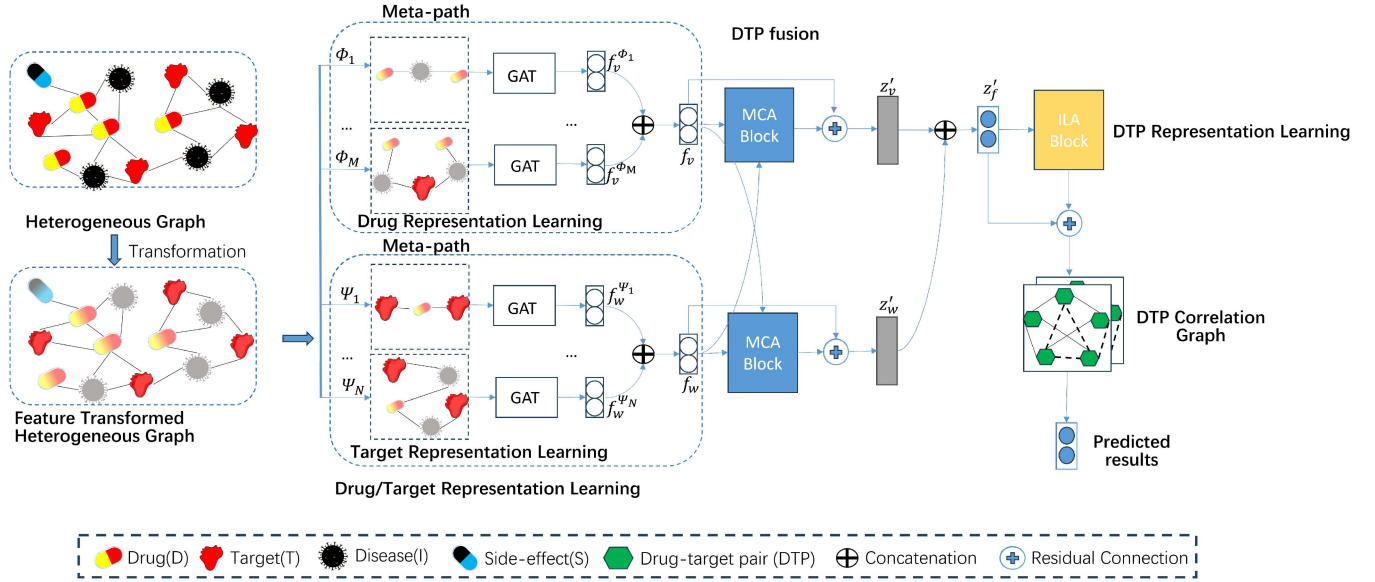


Fig. 1. The overall framework of CAIHGNN. There are M metapaths for drug representation learning and N metapaths for target representation learning, respectively. Each metapath specifies biological semantic relations in a certain substructure. MCA block stands for Multi-Head Cross Attention block. ILA block refers to Intra-layer Attention block. In DTP correlation graph, the real lines and the dotted lines indicate strong associations and weak associations between DTPs, respectively.

meta-paths for drug nodes and target nodes separately. For drug nodes, both the starting and ending points of the meta-path are drugs, and the same applies to target nodes. The maximum length of the meta-path is limited to 5, which is semantically sufficient to capture structural information. Additionally, excessively long meta-paths not only increase computational complexity and memory usage but also introduce misleading information. Thus, we established ten types of meta-paths for drug nodes (i.e., DD, DTD, DID, DSD, DTTD, DTTDD, DIDID, DSDSD, DTITD, DTITD) and eight types of meta-paths for target nodes (i.e., TT, TDT, TIT, TDDT, TITIT, TDTDT, TDIDT, TIDIT). These meta-paths are designed to capture semantic relationships involving disease and side effect nodes, following the framework proposed in [13]. For instance, the DID meta-path (Drug-Disease-Drug) indicates that two drugs may share similar target interactions if they are associated with the same disease, while the DSD meta-path (Drug-Side Effect-Drug) suggests potential similarities in drug-target interactions based on shared side effects. By leveraging these meta-paths, the model incorporates disease and side effect information into the semantic representations of drugs and targets, which are further utilized in subsequent feature integration and prediction stages.

B. Drug/Target Representation Learning

1) *Feature Transformation*: In order to process drugs, targets, diseases, and side effects in the same feature space, we first initialized each node using one-hot encoding, and then applied a feature transformation matrix W_o to linearly transform the original feature vectors.

$$\mathbf{f}'_v = \mathbf{W}_o \cdot \mathbf{x}_v \quad (1)$$

In the formula, $\mathbf{f}'_v$ represents the transformed node vector representation, while $\mathbf{x}_v$ denotes the initial node vector representation. The subscript o refers to the node type. The transformation matrix addresses the issue of original vectors existing in different feature spaces, facilitating subsequent processes.

2) *Message Passing*: We define Φ as a meta-path in a biological heterogeneous graph, from which we can obtain a meta-graph G_v^Φ consisting of numerous instances ϕ_{vu} (where $u \in N_v^\Phi$). Information is transmitted along each meta-path from ϕ_{vu} to the node v .

$$\mathbf{f}'_{\phi_{vu}} = f_\theta(\phi_{vu}) = f_\phi(\mathbf{f}'_i, \forall i \in \phi_{vu}), \quad (2)$$

In this context, $\mathbf{f}'_{\phi_{vu}}$ denotes the encoded representation of the meta-path ϕ_{vu} , while f_θ represents the message transmission function. In this study, the instances of the meta-path encompass all nodes along the path, and the message transmission function is implemented using max-pooling.

3) *Message Update*: Considering that different instances of meta-paths have varying impacts on node v , we chose to utilize Graph Attention Networks (GAT) to update the features of node v . To further enhance the model's expressive capability and stability, we opted to implement a multi-head attention mechanism with K heads, applying self-attention independently on each head and then concatenating the results.

$$\begin{aligned} \mathbf{f}'_v &= \big\|_{k=1}^K \sigma \left(\sum_{u \in N_v^\Phi} [\alpha_{\phi_{vu}}]_k \cdot \mathbf{f}'_{\phi_{vu}} \right), \\ e_{\phi_{vu}} &= \delta(\mathbf{a}_\Phi^T \cdot [\mathbf{f}'_v \| \mathbf{f}'_{\phi_{vu}}]), \\ \alpha_{\phi_{vu}} &= \text{softmax}(e_{\phi_{vu}}) = \frac{\exp(e_{\phi_{vu}})}{\sum_{u' \in N_v^\Phi} \exp(e_{\phi_{vu'}})} \end{aligned} \quad (3)$$

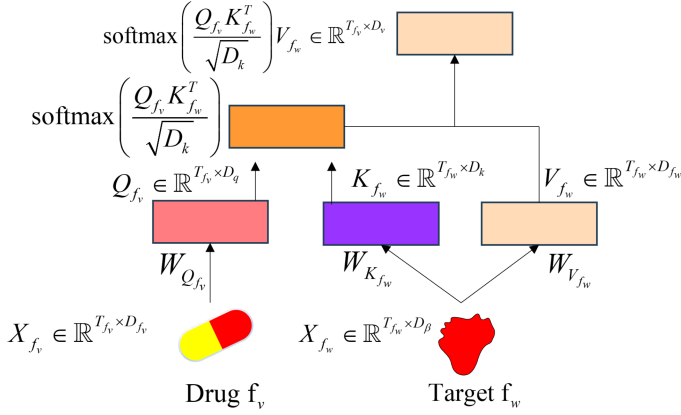


Fig. 2. Architectural elements of a multi-head cross-attention block between drug feature f_v and target feature f_w .

In this section, $\parallel$ denotes the concatenation operation, and $[\alpha_{\phi_{vu}}]_k$ represents the normalized attention coefficient of the meta-path instance ϕ_{vu} for node v under the k -th attention head. α_{Φ}^T is the learnable parameter of the meta-path Φ , $\sigma(\cdot)$ is the Exponential Linear Unit (ELU) function [29], and δ is the Leaky ReLU function.

4) *Feature Integration*: Through the above steps, for each drug $v \in V_D$, there are M meta-paths, $\Omega_D = \{\Phi_1, \Phi_2, \dots, \Phi_M\}$. We obtained the vector representations of the M meta-paths, $\{f_v^{\Phi_1}, f_v^{\Phi_2}, \dots, f_v^{\Phi_M}\}$. Each meta-path represents a semantic aspect embedded in node v . Furthermore, we aggregate this information using the function $g(\cdot)$, where $g(\cdot)$ denotes max-pooling.

$$\mathbf{f}_v = g(\{f_v^{\Phi_1}, f_v^{\Phi_2}, \dots, f_v^{\Phi_M}\}) = \parallel_{m=1}^M f_v^{\Phi_m} \quad (4)$$

Similarly, for the target node $w \in V_T$, there are N meta-paths, $\Omega_T = \{\Psi_1, \Psi_2, \dots, \Psi_N\}$. We also extracted the vector representations of the N meta-paths, $\{f_w^{\Psi_1}, f_w^{\Psi_2}, \dots, f_w^{\Psi_N}\}$. By applying max-pooling, we obtained the final vector representation $\mathbf{f}_w$.

C. DTP Fusion

After completing the feature encoding representations for drugs and targets, we input the matrices $\mathbf{F}_v$ and $\mathbf{F}_w$, composed of B vectors $\mathbf{f}_v$ and $\mathbf{f}_w$, into the DTP fusion module. The DTP fusion module is designed to capture the interaction information between drugs and targets with a Multi-head Cross Attention Block (MCA), shown in Fig. 2. Therefore, in this module, not only is the information of the drugs utilized, but the information of the targets is also incorporated. Specifically, for the drugs, we perform a linear transformation on the input drug features to obtain the queries required for computing the attention output, and we apply a linear transformation to the input target features to obtain the keys and values.

$$\begin{aligned} \mathbf{Q}_h &= \mathbf{F}_v W_{Q_h}, \\ \mathbf{K}_h &= \mathbf{F}_w W_{K_h}, \end{aligned}$$

$$\mathbf{V}_h = \mathbf{F}_w W_{V_h},$$

$$e_{ij}^h = \frac{\mathbf{Q}_{i,h} \cdot \mathbf{K}_{j,h}^T}{\sqrt{d_k}},$$

$$\alpha_{ij}^h = \text{softmax}(e_{ij}^h) = \frac{\exp(e_{ij}^h)}{\sum_{j=1}^n \exp(e_{ij}^h)},$$

$$\mathbf{Z}_{i,h} = \sum_{j=1}^n \alpha_{ij}^h \mathbf{V}_{j,h},$$

$$\mathbf{Z}_i = \parallel_{h=1}^H \mathbf{Z}_{i,h}, \quad (5)$$

Specifically, h represents the h th heads. Then, the obtained result $\mathbf{Z}$ is aggregated with the initial drug information $\mathbf{F}_v$ as additional information.

$$\mathbf{Z}'_v = \mathbf{Z} + \mathbf{F}_v \quad (6)$$

We obtain the drug node representation $\mathbf{Z}'_v$ with additional drug-target association information. Similarly, a similar operation is performed for the target information. Finally, by concatenating the drug information $\mathbf{Z}'_v$ and the target information $\mathbf{Z}'_w$, we obtain the representation for each DTP pair, which is given by $\mathbf{Z}'_f = \mathbf{Z}'_v \parallel \mathbf{Z}'_w$.

The adoption of cross-attention allows the same drug (or target) to have distinct representations in different DTPs by dynamically modeling their interactions, which increases the semantic distance between DTPs in the representation space. This context-dependent representation enhances the model's ability to learn discriminative DTP semantics, crucial for accurate DTI prediction in heterogeneous graphs where DTP interactions are influenced by complex semantic factors.

D. DTP Representation

Considering that similar drugs often interact with similar targets, and vice versa, the association of DTP pairs with similar drugs or targets is stronger than that of DTP pairs without similar drugs or targets. Based on this consideration, we designed two methods to leverage the similarity between DTP pairs. First, as shown in Fig. 3, for each DTP node, we treat it as a query and the other DTP nodes as keys and values, applying a Intra-Layer Attention mechanism (ILA). Given an initial set of vectors $\mathbf{Z}' = \{z'_1, z'_2, \dots, z'_n\}$, where each vector $z'_i \in \mathbb{R}^d$ represents an input embedding. To calculate the intra-layer attention, each vector z'_i acts as the query Q , while all other vectors z'_j serve as the keys K and values V in the attention computation.

- 1) First, project each vector z'_i into query, key, and value spaces:

$$Q_i = z'_i W_Q, \quad K_j = z'_j W_K, \quad V_j = z'_j W_V \quad (7)$$

where W_Q , W_K , and W_V are learnable weight matrices.

- 2) For each query Q_i , calculate the attention weights α_{ij} between Q_i and each key K_j using scaled dot-product

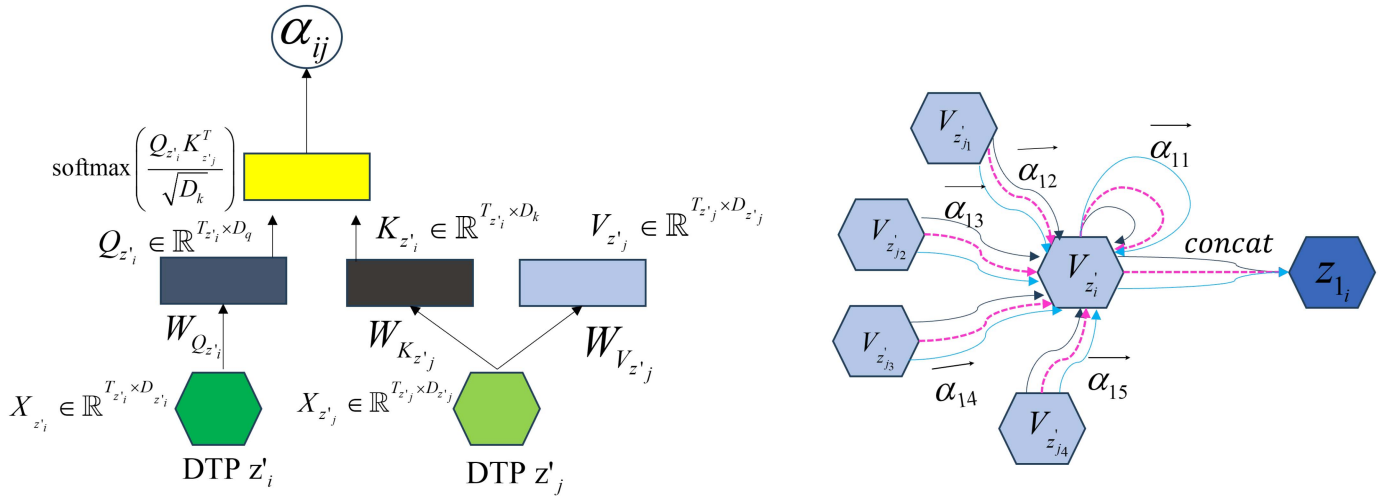
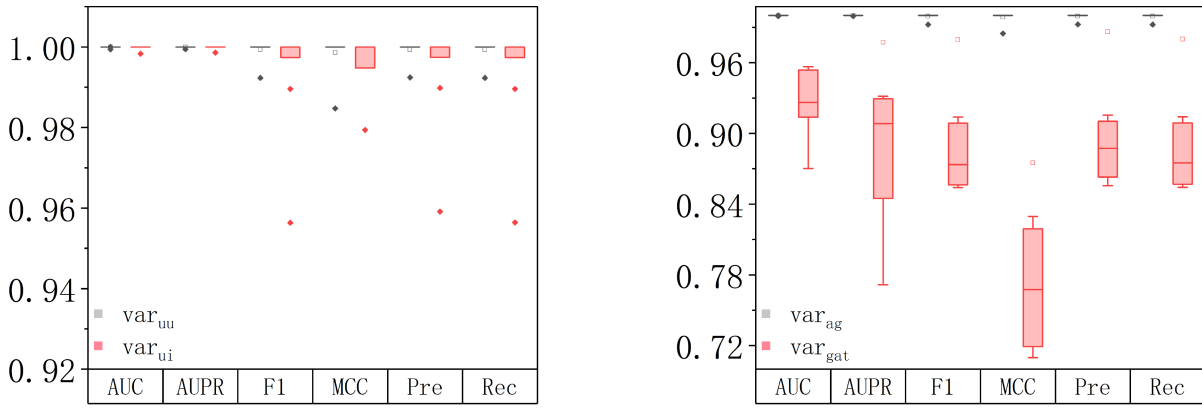


Fig. 3. Architectural Elements of an Intra-Layer-Attention Block Between DTP node z'_i and other DTP nodes z'_j . **Left:** The method for obtaining the weights α_{ij} . **Right:** An illustration of multi-head attention (with $K=8$ heads) by node z'_i on its neighborhood. Different arrow styles and colors denote independent attention computations. The aggregated features from each head are concatenated to obtain z_{1_i} .



(a) Attention analyze on dataset Hetero-A between var_{uu} and var_{ui} (b) Attention analyze on dataset Hetero-A between var_{ag} and var_{gat}

Fig. 4. Attention analyze on dataset Hetero-A.

attention:

$$\alpha_{ij} = \frac{\exp\left(\frac{Q_i K_j^T}{\sqrt{d_k}}\right)}{\sum_{j=1}^n \exp\left(\frac{Q_i K_j^T}{\sqrt{d_k}}\right)} \quad (8)$$

where d_k is the dimension of the key vectors, used for scaling.

- 3) The output z_{1_i} for each query Q_i is then a weighted sum of the value vectors V_j :

$$z_{1_i} = \sum_{j=1}^n \alpha_{ij} V_j \quad (9)$$

- 4) Aggregate all outputs z_{1_i} to form the matrix $Z_1 = \{z_{1_1}, z_{1_2}, \dots, z_{1_n}\}$.

- 5) Finally, add a residual connection by summing Z_1 and the original input Z' element-wise to produce the final output Z_2 :

$$Z_2 = Z_1 + Z' \quad (10)$$

This final matrix Z_2 represents the Intra-layer attention output with a residual connection, enabling enhanced information flow and gradient propagation.

Without intra-layer attention, the fused DTP representation would treat all features equally, potentially overlooking the nuanced contributions of less similar semantic components. By applying intra-layer attention, the model can better capture the information from diverse semantic aspects within the DTP representation, even those that are less similar in the semantic

space, leading to a more comprehensive representation for DTI prediction.

E. DTI Prediction

Intra-layer attention allows us to learn the interaction information between DTP nodes. We then add this additional information to the original DTP node, resulting in a DTP node enriched with additional interaction information. Next, we establish a DTP correlation graph $G_{DTP} = (\mathbf{Z}_2, \mathbf{A})$ to further explore the latent information between DTP pairs. The adjacency matrix $\mathbf{A}$ can be obtained from the vector representations of the DTPs.

$$\mathbf{A} = \text{softmax}(\mathbf{Z}_2 \mathbf{Z}_2^T) \quad (11)$$

In Formula (7), $\mathbf{A}$ is a symmetric matrix, where A_{ij} represents the normalized similarity score between the i -th DTP and the j -th DTP. We then input $\mathbf{Z}_2$ into the GCN, and the result of the second layer of the GCN is the model's predicted outcome.

$$\mathbf{y} = \mathbf{A}(\text{ReLU}(\mathbf{A}\mathbf{Z}_2\mathbf{W}_1))\mathbf{W}_2, \quad (12)$$

where $\mathbf{W}_1$ and $\mathbf{W}_2$ are learnable parameters that aggregate the information of the DTP nodes themselves, compressing information from a high-dimensional space into a low-dimensional space. The matrix $\mathbf{A}$ is responsible for explicitly aggregating graph information through the semantic similarity between its own nodes and neighboring nodes. Considering this is a binary classification problem, we decided to use the cross-entropy function as the loss function to evaluate the gap between the predicted values and the true values. Its formula is as follows:

$$L = -\frac{1}{B} \sum_{b=1}^B [y_b \log(y'_b) + (1 - y_b) \log(1 - y'_b)] \quad (13)$$

where B is the number of DTPs trained in a batch. If the b -th DTP has an interaction, then $y_b = 1$; otherwise, $y_b = 0$. The term y'_b represents the predicted probability of interaction for the b -th DTP after normalization using the softmax function.

IV. EXPERIMENT

A. Dataset

To evaluate the performance of our model on DTI prediction, we conducted experiments on two biological heterogeneous datasets. The first dataset (referred to as Hetero-A) was introduced in DTINet [30]. It consists of 708 drug nodes, 1512 target nodes, 5603 disease nodes, and 4192 side effect nodes, including six types of connections: D-T, D-D, D-I, D-S, T-T, and T-I. The second dataset (referred to as Hetero-B) was introduced in MHGNN [13]. It consists of 2214 drug nodes, 1968 target nodes, 7205 disease nodes, and 3935 side effect nodes, also including six types of connections: D-T, D-D, D-I, D-S, T-T, and T-I. Additional results for Hetero-B, including tables and figures, are provided in Appendix.

B. Evaluation Metric

In this study, we employed six evaluation metrics to comprehensively assess the performance of our model: Precision

Score, Recall Score, F1 Score, Matthews Correlation Coefficient (MCC), Area Under the Curve (AUC), and Area Under the Precision-Recall Curve (AUPR). These metrics provide a multifaceted view of the model's effectiveness in a binary classification context.

To compute these metrics, we first determined the following values from the confusion matrix: True Negatives (TN), False Negatives (FN), True Positives (TP), and False Positives (FP).

- **Precision Score:** Precision measures the proportion of true positive predictions among all positive predictions made by the model. It is calculated as:

$$\text{Precision} = \frac{\text{TP}}{\text{TP} + \text{FP}} \quad (14)$$

- **Recall Score:** Recall indicates the proportion of actual positive instances that were correctly identified by the model. Its calculation is given by:

$$\text{Recall} = \frac{\text{TP}}{\text{TP} + \text{FN}} \quad (15)$$

- **F1 Score:** The F1 Score is the harmonic mean of Precision and Recall, providing a single metric that balances both concerns, particularly in cases of class imbalance. It is defined as:

$$\text{F1 Score} = 2 \times \frac{\text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}} \quad (16)$$

- **Matthews Correlation Coefficient (MCC):** The MCC is a robust metric that takes into account all four quadrants of the confusion matrix. It is particularly useful for imbalanced datasets and is calculated as:

$$\text{MCC} = \frac{(\text{TP} \times \text{TN}) - (\text{FP} \times \text{FN})}{\sqrt{(\text{TP} + \text{FP})(\text{TP} + \text{FN})(\text{TN} + \text{FP})(\text{TN} + \text{FN})}} \quad (17)$$

- **Area Under the Curve (AUC):** AUC measures the ability of the model to distinguish between positive and negative classes. It is derived from the Receiver Operating Characteristic (ROC) curve and ranges from 0 to 1, with higher values indicating better performance.
- **Area Under the Precision-Recall Curve (AUPR):** AUPR evaluates the trade-off between precision and recall for different threshold values. It is considered to better reflect a model's ability to enrich positive samples compared to AUC.

These metrics were selected because they collectively provide a comprehensive evaluation of the model's performance, allowing us to analyze its strengths and weaknesses in various aspects of classification tasks.

C. Implementation Detail

In our model, we applied 10-fold cross-validation, where the training and evaluation processes were independently conducted 10 times, including the meta-path construction process. All known DTIs were used as positive samples, and the same number

of randomly selected unknown drug–target pairs were used as negative samples. For each fold, 90% of the positive and negative samples were randomly selected as the training set, with the remaining 10% used as the test set. We adopted the Adam optimizer with a weight decay rate of 1×10^{-5} , and the learning rate was selected from $\{1 \times 10^{-5}, 1 \times 10^{-4}\}$. The model was trained for 250 epochs, using an early stopping strategy. The projection and hidden feature dimensions were set to 64, the batch size was 32, and the number of attention heads k was set to 8.

D. Baseline

We compared our model with 18 baseline methods, including 5 similarity-based algorithms, 6 knowledge graph-based algorithms, 4 graph neural network-based algorithms, and 3 heterogeneous graph embedding methods. These baselines were carefully selected to represent the latest advancements in their respective categories, covering a wide range of state-of-the-art techniques for drug-target interaction (DTI) prediction.

1) Similarity-based algorithms:

- *DTINet* [30] utilizes random walk with restart (RWR) to extract representations from drug and target similarity networks, followed by inductive matrix completion (IMC) for DTI prediction.
- *deepDTnet* [31] converts drug and target similarity networks into representation matrices using RWR and positive pointwise mutual information (PPMI), and learns features using an autoencoder (AE) model.
- *MEDTI* [32] is similar to deepDTnet but combines multiple representation matrices and learns features through an autoencoder.
- *NEDTP* [33] extracts features using Word2Vec and predicts using gradient boosting decision tree (GBDT).
- *MultiDTI* [34] extracts representations from sequences using convolutional layers with residual connections and maps them into a common space.

2) Knowledge graph-based methods:

- *TransE* [35], *DistMult* [36], and *ComplEx* [37] are knowledge graph embedding methods that derive low-dimensional representations of bioentities for DTI prediction. These methods employ translation-based, bilinear, and complex-valued approaches, respectively.
- *KGE_NFM* [38], *TransE-NFM*, and *ComplEx-NFM* enhance these embeddings by integrating neural factorization machines (NFM) and feedforward neural networks. They adapt DistMult, TransE, and ComplEx, respectively, to improve DTI prediction accuracy.

3) Graph neural network-based methods:

- *NeoDTI* [39] integrates representations using GCNs and performs prediction with IMC.
- *IMCHGAN* [40] employs a two-level attention mechanism to learn features and predicts using IMC.
- *DTI-MGNN* [41] and *SGCL-DTI* [14] extract representations from meta-paths using GCN and optimize using a contrastive learning strategy.

4) Heterogeneous graph embedding methods:

- *HAN* [42] utilizes meta-paths and attention mechanisms for feature fusion, customized for DTI prediction.
- *MAGNN* [43] is an extension of HAN, which considers all node information for feature fusion and DTI prediction.
- *MHGNN* [13] uses meta-paths for dual-channel learning and GCN for DTI prediction.

E. Performance Comparison

For fair comparison, all these baselines were trained and tested on the same datasets as our model. For each evaluation metric, we calculated both the average value and the standard deviation. The comparison results are shown in Table I. It can be observed that in Hetero-A, CAIHGNN consistently outperforms all other methods, with improvements in metrics such as AUC, AUPR, precision score, recall score, F1 score, and MCC of 0.0272, 0.0447, 0.0323, 0.0037, 0.0333, and 0.0673, respectively. In Hetero-B, CAIHGNN exhibits the best performance in five of the six metrics, with the remaining metric showing little difference from the baseline. The results demonstrate the superiority of CAIHGNN in DTI prediction. Compared to similarity-based methods, CAIHGNN is an end-to-end structure that can capture high-order relationships in the biological heterogeneous graph. In contrast, similarity-based methods exhibit a simpler approach in feature learning for drugs and targets, capturing only low-order relationships within the heterogeneous graph. This limitation contributes to the inferior performance of similarity-based methods in DTI prediction compared to CAIHGNN. Compared to KG-based methods, CAIHGNN naturally represents multiple types of nodes and relationships, while KG-based methods require extensive biomedical knowledge, making the process cumbersome and prone to limitations based on the accuracy and completeness of the data. Additionally, different types of relationships in the knowledge graph may contribute differently to the prediction task, and simple feature extraction may not adequately capture these differences. Similar to our work, MHGNN utilizes metapaths and GCN to predict DTIs. Nevertheless, it does not consider the interrelationships between drugs and targets during feature learning; instead, it directly concatenates them. This results in ineffective mutual recognition between drugs and targets. In contrast, in CAIHGNN, we first perform cross attention on the learned features of drugs and targets, integrating their information and enhancing the feature representation of drug-target pairs (DTPs).

Furthermore, MHGNN directly uses GCN at the decision layer to predict whether there is an association between drug-target pairs (DTPs). This approach is based on aggregating relationships after the initial learning of drugs and targets, implying that the importance of different neighbors in the DTP graph is determined by similarity. In contrast, we first employ an attention mechanism in the DTP graph to further adjust the similarities between neighbors, and then utilize GCN for prediction, thereby enhancing the aggregation of neighbor information within the DTP graph.

F. Attention Analyze

To validate the rationality of cross-attention and the attention mechanism between DTI nodes, we conduct experiments on the

TABLE I
COMPARISON OF CAIHGNN WITH 17 METHODS ON DATASET HETERO-A. THE HIGHEST AND THE SECOND HIGHEST RESULTS OVER EACH MEASUREMENT ARE IN BOLD AND UNDERLINED, RESPECTIVELY.

Methods	AUC $\pm$ std	AUPR $\pm$ std	Precision $\pm$ std	Recall $\pm$ std	F1 $\pm$ std	MCC $\pm$ std
DTNet	0.8838 $\pm$ 0.022	0.9024 $\pm$ 0.017	0.8677 $\pm$ 0.024	0.7369 $\pm$ 0.024	0.7967 $\pm$ 0.020	0.6316 $\pm$ 0.036
deepDTNet	0.7688 $\pm$ 0.040	0.7744 $\pm$ 0.040	0.6970 $\pm$ 0.031	0.6938 $\pm$ 0.044	0.6952 $\pm$ 0.036	0.3928 $\pm$ 0.065
MEDTI	0.7421 $\pm$ 0.038	0.7428 $\pm$ 0.029	0.6796 $\pm$ 0.030	0.7036 $\pm$ 0.038	0.6913 $\pm$ 0.033	0.3726 $\pm$ 0.064
NEDTP	0.9115 $\pm$ 0.021	0.919 $\pm$ 0.021	0.8698 $\pm$ 0.023	0.7993 $\pm$ 0.037	0.8328 $\pm$ 0.029	0.6822 $\pm$ 0.050
MultiDTI	0.8951 $\pm$ 0.016	0.9126 $\pm$ 0.015	0.8253 $\pm$ 0.127	0.7882 $\pm$ 0.106	0.7898 $\pm$ 0.031	0.6007 $\pm$ 0.098
TransE	0.7435 $\pm$ 0.035	0.7974 $\pm$ 0.025	0.6496 $\pm$ 0.042	0.7051 $\pm$ 0.035	0.6759 $\pm$ 0.036	0.3242 $\pm$ 0.082
TransE-NFM	0.9247 $\pm$ 0.014	0.9286 $\pm$ 0.014	0.8770 $\pm$ 0.019	0.8201 $\pm$ 0.024	0.8473 $\pm$ 0.016	0.7065 $\pm$ 0.030
DistMult	0.7436 $\pm$ 0.035	0.7975 $\pm$ 0.025	0.6492 $\pm$ 0.043	0.7046 $\pm$ 0.033	0.6754 $\pm$ 0.036	0.3232 $\pm$ 0.083
KGE_NFM	0.9250 $\pm$ 0.014	0.9285 $\pm$ 0.015	0.8839 $\pm$ 0.025	0.8164 $\pm$ 0.026	0.8484 $\pm$ 0.017	0.7110 $\pm$ 0.032
ComplEx	0.7436 $\pm$ 0.035	0.7977 $\pm$ 0.025	0.6515 $\pm$ 0.042	0.7051 $\pm$ 0.033	0.6769 $\pm$ 0.036	0.3272 $\pm$ 0.081
ComplEx-NFM	0.9242 $\pm$ 0.013	0.9265 $\pm$ 0.017	0.8784 $\pm$ 0.019	0.8180 $\pm$ 0.029	0.8469 $\pm$ 0.021	0.7066 $\pm$ 0.039
NeoDTI	0.8690 $\pm$ 0.052	0.8821 $\pm$ 0.047	0.7959 $\pm$ 0.074	0.7904 $\pm$ 0.038	0.7918 $\pm$ 0.050	0.5825 $\pm$ 0.116
IMCHGAN	0.8862 $\pm$ 0.021	0.9000 $\pm$ 0.021	0.8747 $\pm$ 0.023	0.7349 $\pm$ 0.046	0.7982 $\pm$ 0.033	0.6384 $\pm$ 0.053
DTI-MGNN	0.8969 $\pm$ 0.026	0.8940 $\pm$ 0.025	0.8725 $\pm$ 0.028	0.8133 $\pm$ 0.027	0.8418 $\pm$ 0.025	0.6960 $\pm$ 0.049
SGCL-DTI	0.7462 $\pm$ 0.108	0.7910 $\pm$ 0.099	0.7415 $\pm$ 0.097	0.628 $\pm$ 0.133	0.6768 $\pm$ 0.115	0.4193 $\pm$ 0.186
HAN	0.9484 $\pm$ 0.003	0.9478 $\pm$ 0.002	0.7902 $\pm$ 0.021	0.9271 $\pm$ 0.010	0.8528 $\pm$ 0.008	0.6908 $\pm$ 0.018
MAGNN	0.9531 $\pm$ 0.011	0.9410 $\pm$ 0.018	0.9469 $\pm$ 0.025	0.5376 $\pm$ 0.090	0.6803 $\pm$ 0.081	0.5619 $\pm$ 0.066
MHGNN	0.9727 $\pm$ 0.011	0.9552 $\pm$ 0.026	0.9316 $\pm$ 0.021	0.9886 $\pm$ 0.009	0.9590 $\pm$ 0.009	0.9174 $\pm$ 0.019
CAIHGNN	0.9999$\pm$0.001	0.9999$\pm$0.001	0.9792$\pm$0.068	0.9923$\pm$0.125	0.9923$\pm$0.151	0.9847$\pm$0.210

TABLE II
ATTENTION ANALYZE ON DATASET HETERO-A

Groups	AUC $\pm$ std	AUPR $\pm$ std	Precision $\pm$ std	Recall $\pm$ std	F1 $\pm$ std	MCC $\pm$ std
Var_{uu}	0.9999 $\pm$ 0.001	0.9999 $\pm$ 0.001	0.9792 $\pm$ 0.068	0.9923 $\pm$ 0.125	0.9923 $\pm$ 0.151	0.9847 $\pm$ 0.209
Var_{ui}	0.9999 $\pm$ 0.001	0.9999 $\pm$ 0.001	0.9815 $\pm$ 0.008	0.9809 $\pm$ 0.008	0.9924 $\pm$ 0.002	0.9984 $\pm$ 0.004
Var_{ag}	0.9999 $\pm$ 0.001	0.9999 $\pm$ 0.001	0.9792 $\pm$ 0.068	0.9923 $\pm$ 0.125	0.9923 $\pm$ 0.151	0.9847 $\pm$ 0.209
Var_{gat}	0.9229 $\pm$ 0.026	0.8750 $\pm$ 0.057	0.8847 $\pm$ 0.021	0.8778 $\pm$ 0.021	0.8773 $\pm$ 0.021	0.7625 $\pm$ 0.041

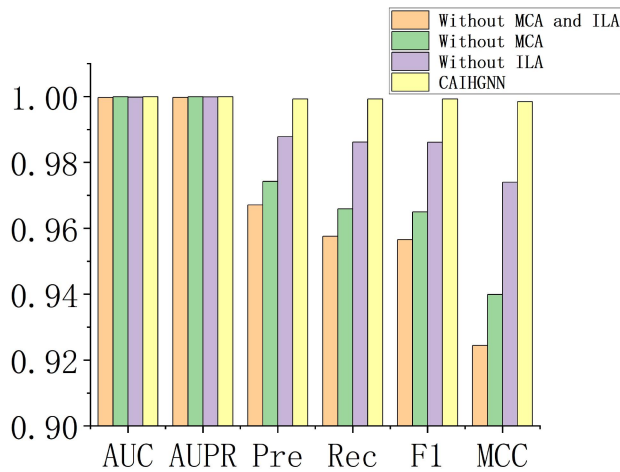
following variants. Var_{uu} considers attaching the result of using drug as the query and target as the key and value to the drug, while also attaching the result of using target as the query and drug as the key and value to the target. Var_{ui} considers attaching the result of using target as the query and drug as the key and value to the drug during cross-attention, while also attaching the result of using drug as the query and target as the key and value to the target. Var_{ag} considers directly applying GCN learning on the graph after completing the attention mechanism between DTI nodes. Var_{gat} represents the direct application of GAT learning on the graph. As shown in Fig. ??, Var_{uu} and Var_{ui} exhibit little difference in performance, indicating that during cross-attention, the query being either drug or target does not significantly affect the additional information associated with drug (target). A more detailed observation of the data can be found in Table II and Fig 4. Var_{ag} performs better than Var_{gat} on both Hetero-A and Hetero-B. This is because Var_{gat} only uses attention in its prediction phase without leveraging the learned features of drugs and targets, which leads to neglect of crucial neighborhood structure information from the graph. In contrast, Var_{ag} not only utilizes the neighborhood structure of nodes but also applies a secondary adjustment to the weights of neighboring information. Consequently, Var_{ag} outperforms Var_{gat} , which indicates that the predictive capability of gcnn + Intra-layer attention is stronger than that of GAT alone.

G. Ablation Study

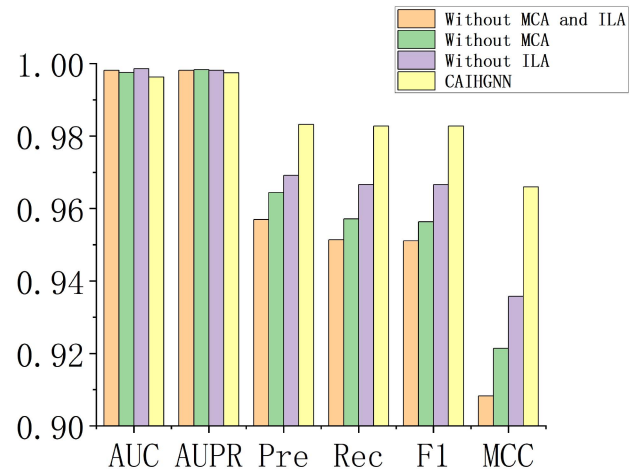
In order to analyze the impact of each component in our model, we conducted an ablation study on the Hetero-A and Hetero-B datasets. The objective of this study is to validate the contribution of each component by comparing the performance of the full model with that of models that lack certain parts. The baseline model, referred to as “CAIHGNN,” includes all components: meta-path aggregation, cross attention, intra-layer attention, and GCN for prediction. For comparison, we designed several ablated models:

- *Without Cross Attention and Intra-layer Attention:* This variant removes both cross attention and intra-layer attention, while retaining meta-path aggregation and GCN for prediction.
- *Without Cross Attention:* Here, cross attention is replaced with simple concatenation during DTP pair aggregation.
- *Without Intra-layer Attention:* In this model, neighbor weights are not redistributed prior to GCN-based prediction.

As shown in Fig. 5(a) and (b), our model achieves the best performance across multiple metrics on both datasets, demonstrating its overall superiority compared to other models. The model “Without Intra-layer Attention” performs slightly worse but still outperforms the “Without Cross Attention” model, while all models surpass the “Without Cross Attention and Intra-layer

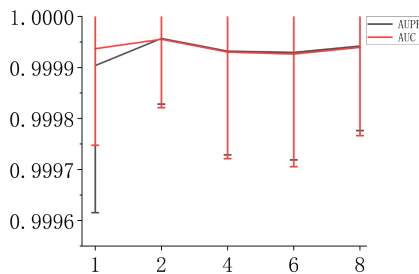


(a) Ablation studies on dataset Hetero-A

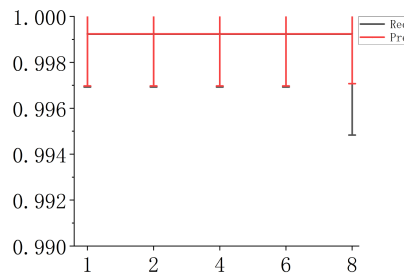


(b) Ablation studies on dataset Hetero-B

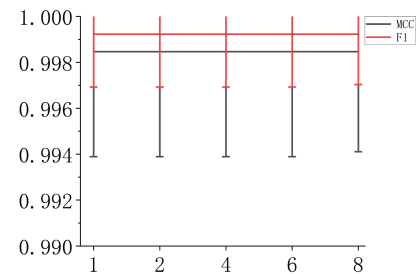
Fig. 5. Ablation studies for MCA and ILA on dataset (a) Hetero-A and (b) Hetero-B.



(a) AUC&AUPR



(b) Rec&Pre



(c) F1&MCC

Fig. 6. Impacts of different number of attention heads on Hetero-A.

Attention” variant. These results lead us to the following conclusions:

- i) Cross attention enhances the feature representation of DTP pairs by enabling mutual recognition between drugs and targets, which proves more effective than simple concatenation of drug and target features.
- ii) Intra-layer Attention further improves GCN predictive performance by fine-tuning the weight distribution among neighboring nodes.

H. Parameter Analysis

We investigate the performances of CAIHGNN with different parameter settings and report the results on Hetero-A and Hetero-B in figures from Fig. 6 to Fig. ??.

1) *Number of Attention Heads K*: We first explore the performance of CAIHGNN with different numbers of attention heads. According to the results in Fig. 6, we find that for Hetero-A, the performance of CAIHGNN shows almost no variation in the AUC score, AUPR score, precision score, recall score, F1-score, and MCC, indicating that CAIHGNN Hetero-A performs excellently and exhibits good robustness. In contrast, for Hetero-B,

CAIHGNN achieves the highest results across these metrics with 6 attention heads, while it attains the lowest results with 1 attention head. These results indicate that an increased number of attention heads enhances model robustness and expressiveness. However, beyond a certain number, adding more attention heads may lead to the risk of overfitting, thereby reducing the model’s performance capability.

2) *Dimension of Embeddings*: The projected and hidden embeddings are set to the same dimension. We vary the dimension from 32 to 256 to analyze DTI prediction performances. The results over different metrics on the Hetero-A dataset are illustrated in Fig. 7, and those on the Hetero-B dataset are presented in Fig. ?. From Fig. 7, we can observe that the model performs similarly when the dimension of embeddings is 32, 64, or 128; however, the performance is inferior when the dimension of embeddings is 256. In contrast, from Fig. ?, we see that the model exhibits relatively stable performance across different dimensions of embeddings, with the best performance achieved at an embedding dimension of 64. This indicates that the model demonstrates good robustness regarding embeddings, but the dimensionality of embeddings should not be too high, as it may lead to model overfitting.

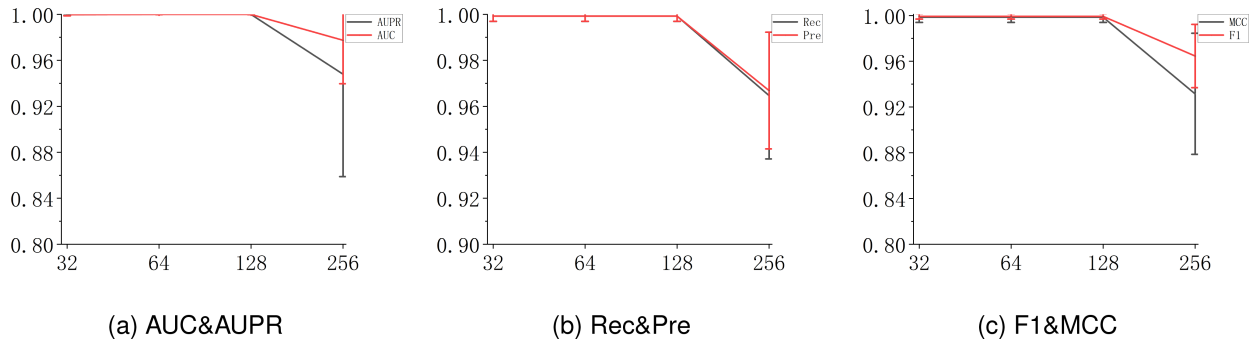


Fig. 7. Impacts of dimension of embeddings on Hetero-A.

TABLE III
COLD-START STUDIES ON DATASETS HETERO-A

No	AUC	AUPR	Precision	Recall	F1	MCC
0	0.9914	0.9928	0.9437	0.9358	0.9355	0.8795
3	0.9875	0.9905	0.9579	0.9540	0.9539	0.9119
5	0.9808	0.9890	0.9631	0.9600	0.9599	0.9231

I. Cold-Start Analysis

Previous experiments demonstrate that our model performs well in DTI prediction; however, in real-life situations,

We often have limited knowledge about the interactions between drugs and targets, which can significantly reduce the model’s generalization capability. To evaluate the robustness of CAIHGNN in sparse scenarios, we conduct a cold-start analysis following the setup of MHGNN [13]. In this analysis, we define the “relationships” of a drug-target pair (DTP) as follows: two DTPs are considered to have a relationship (i.e., an edge between them) if they share the same drug or the same target. Based on this definition, we select DTPs with no more than L relationships as cold-start ones, with $L \in \{0, 3, 5\}$ for Hetero-A (fewer DTPs, lower density) and $L \in \{3, 5, 8\}$ for Hetero-B (more DTPs, higher density) to reflect dataset characteristics and simulate sparsity. The results are shown in Tables III and ?. We can observe from Table III that No.5 outperforms No.3 and No.0 in four metrics, yet No.0 shows the best performance in AUC and AUPR. This may be due to the fact that, when relationships are limited to zero, the number of valid DTPs decreases significantly (only a few dozen, whereas the original dataset contains thousands), leading to biased results that inflate AUC and AUPR values. Overall, however, we find that the model’s performance improves as the number of relationships increases, indicating the stability of the model structure. In Table ?, the metrics of No.5 and No.8 are close and both outperform No.3, indicating that the model achieves stability when relationships are set to 5. This is likely because, at this level, most DTPs are already included.

J. Generalization Analysis With Cross-Dataset Validation

To evaluate CAIHGNN’s generalization in real-world scenarios with unseen drug-target interactions, we perform cross-dataset validation. We train on Hetero-A and test on Hetero-B

(A ON B), and vice versa (B ON A). Due to differing embedding layer dimensions, we replace the embedding layer of the model trained on Hetero-A with Hetero-B’s, perform minimal fine-tuning, and test on Hetero-B (and vice versa). This setup tests the model’s robustness across datasets, with results in Table IV.

As shown in the table, CAIHGNN achieves an average AUC of 0.8907 ± 0.0288 , AUPR of 0.8292 ± 0.0611 , Precision of 0.9300 ± 0.0000 , Recall of 0.9150 ± 0.0050 , F1-score of 0.9150 ± 0.0050 , and MCC of 0.8450 ± 0.0050 when trained on Hetero-A and tested on Hetero-B. Conversely, when trained on Hetero-B and tested on Hetero-A, it attains an AUC of 0.9938 ± 0.0043 , AUPR of 0.9928 ± 0.0054 , Precision of 0.9750 ± 0.0150 , Recall of 0.9750 ± 0.0150 , F1-score of 0.9750 ± 0.0150 , and MCC of 0.9500 ± 0.0300 . These results highlight CAIHGNN’s strong generalization capability, especially on Hetero-A. Overall, this cross-dataset validation demonstrates the model’s ability to handle unseen drugs and targets in real-world DTI prediction scenarios.

K. Case Study

The original datasets contain homologous targets and similar drugs, which may cause inflated evaluation performance. To investigate the model’s robustness, we further conduct tests by removing positive DTIs with high homology from the training set and then predicting them in the test phase. Following the approach in [30], we removed high homology DTPs under the following five conditions: (i) Removal of DTPs involving homologous proteins (sequence identity scores $> 40\%$); (ii) Removal of DTPs with similar drugs (Tanimoto coefficients $> 60\%$); (iii) Removal of DTPs where drugs have similar side effects (Jaccard similarity scores $> 60\%$); (iv) Removal of DTPs where drugs or proteins are associated with similar diseases (Jaccard similarity scores $> 60\%$); and (v) Removal of DTPs with similar drugs (Tanimoto coefficients $> 60\%$) or homologous proteins (sequence identity scores $> 40\%$). Tests on Hetero-A (Table V) demonstrate that our model is robust against the removal of homologous proteins or similar drugs in the training data.

L. Metapath Importance Analysis

To enhance the interpretability of CAIHGNN and understand which semantic relationships contribute most to DTI

TABLE IV
PERFORMANCE OF CAIHGNN IN CROSS-DATASET VALIDATION

Setting	AUC	AUPR	Precision	Recall	F1	MCC
A ON B	0.8907±0.0288	0.8292±0.0611	0.9300±0.0000	0.9150±0.0050	0.9150±0.0050	0.8450±0.0050
B ON A	0.9938±0.0043	0.9928±0.0054	0.9750±0.0150	0.9750±0.0150	0.9750±0.0150	0.9500±0.0300

TABLE V
CASE STUDIES ON DATASET HETERO-A

Case	AUC	AUPR	Precision	Recall	F1	MCC
C1	0.9986	0.9989	0.9889	0.9886	0.9886	0.9776
C2	0.9982	0.9982	0.9629	0.9596	0.9595	0.9225
C3	0.9932	0.9815	0.9620	0.9571	0.9557	0.9191
C4	0.9576	0.9115	0.9516	0.9456	0.9454	0.8967
C5	0.9962	0.9939	0.9856	0.9849	0.9849	0.9705

TABLE VI
IMPORTANCE OF DRUG-RELATED AND TARGET-RELATED METAPATHS BASED ON SOFTMAX SCORES (IN PERCENTAGE)

Metapath	Type	Importance (%)
D-D	Drug	24.45
D-T-D-T-D	Drug	20.07
D-T-D	Drug	17.73
D-T-T-D	Drug	9.19
D-T-I-T-D	Drug	8.69
D-I-D	Drug	5.32
D-I-T-I-D	Drug	3.99
D-I-D-I-D	Drug	3.83
D-S-D	Drug	3.67
D-S-D-S-D	Drug	3.06
T-D-I-D-T	Target	33.23
T-D-T-D-T	Target	23.83
T-D-D-T	Target	17.82
T-D-T	Target	14.58
T-T	Target	3.82
T-I-T	Target	2.40
T-I-T-I-T	Target	2.19
T-I-D-I-T	Target	2.12

predictions, we conduct a metapath importance analysis. The importance of each metapath is quantified using a softmax function applied to the learned attention weights during training, reflecting their relative contribution to the final prediction. The results are summarized in Table VI. For drug-related metapaths, the top three most important paths are D-D (24.45%), D-T-D-T-D (20.07%), and D-T-D (17.73%), collectively accounting for approximately 62.45% of the total importance, while paths involving diseases (e.g., D-I-D: 5.32%) and side effects (e.g., D-S-D: 3.67%) contribute less. For target-related metapaths, the top three paths are T-D-I-D-T (33.23%), T-D-T-D-T (23.83%), and T-D-D-T (17.82%), which together account for about 75.03% of the total importance, indicating a strong reliance on drug- and disease-mediated relationships.

The metapath importance analysis reveals key insights into CAIHGNN's predictive behavior. It is noted that for drug-related metapaths, the paths containing only Drug and Target nodes exhibit the highest importance. Similarly, for the top four most important target-related metapaths, they also tend to include

Drug and Target nodes as much as possible. This indicates that CAIHGNN prioritizes direct drug-target interactions when learning drug (target) representations. Among them, the prominent importance of D-D likely reflects the strong predictive power of chemical structure similarity, as drugs with similar structures often target similar proteins. This finding suggests that the metapath learning approach aligns with traditional similarity-based DTI prediction methods. However, traditional methods rely on the similarity of drug chemical structures and target sequences, capturing only simple similarity relationships, whereas CAIHGNN leverages metapaths to capture higher-order, multi-hop drug-target interactions, thus surpassing traditional methods in performance.

V. CONCLUSION

In this paper, we have proposed CAIHGNN to capture complex structures and rich semantics in biological heterogeneous graphs for drug-target interaction (DTI) prediction. CAIHGNN employs a dual-channel architecture to learn drug and target representations, which are fused into DTP representations through cross-attention. A DTP correlation graph is then constructed to exploit high-order correlations between DTPs, with intra-layer attention applied to capture their implicit associations. Comprehensive experiments on two biological heterogeneous datasets validate the effectiveness of CAIHGNN for DTI prediction. Despite these advancements, CAIHGNN has certain limitations: the cross-attention and intra-layer attention mechanisms introduce high computational complexity, limiting scalability on large-scale heterogeneous graphs, and its metapath-based approach relies on auxiliary nodes (e.g., diseases, side effects), potentially leading to degraded performance on sparse or incomplete datasets. For future work, we plan to explore lightweight attention mechanisms or graph sampling techniques to reduce computational overhead, develop adaptive metapath selection strategies by incorporating external knowledge graphs (e.g., DrugBank, KEGG) to enhance robustness, and further mine topological similarities in the DTP graph (e.g., whether DTP pairs share the same drug or target) beyond the current focus on semantic similarity.

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